

PROLOG ASSIGNMENT

The Philosopher and Scholars Problem

You are given the following facts:

1. All philosophers are thinkers.
2. All thinkers are scholars.
3. All scholars are knowledgeable.
4. Aristotle is a philosopher.

You have to prove that:

Aristotle is knowledgeable

Using the Resolution Refutation method in First-Order Predicate Logic (FOL).

Instructions:

You must perform each transformation step of the reasoning process using Prolog. Create separate sections (or predicates) in Prolog that demonstrate every stage—from English sentences to automated resolution.

Show each resolution step and end with the empty clause ($\perp$) proving that the goal follows from the given facts.

Expected Prolog File Structure:

Your .pl file should clearly demonstrate each phase:

```
% Step 1: Predicate logic representation
% Step 2: Negation of goal
% Step 3: Implication removal
% Step 4: Negation movement
% Step 5: Standardization
% Step 6: Skolemization
% Step 7: CNF clauses
% Step 8: Resolution implementation and execution
```

Deliverables:

1. A single Prolog file (philosopher_scholars.pl) containing:
 - The code and comments for all transformation steps.
 - The resolution refutation process showing the derivation of the empty clause.

2. A screenshot of the final proof output (showing that the empty clause or contradiction has been derived).

Note:

This problem is slightly more challenging than the basic three-statement version because the proof requires chaining three universal rules before reaching the final conclusion.